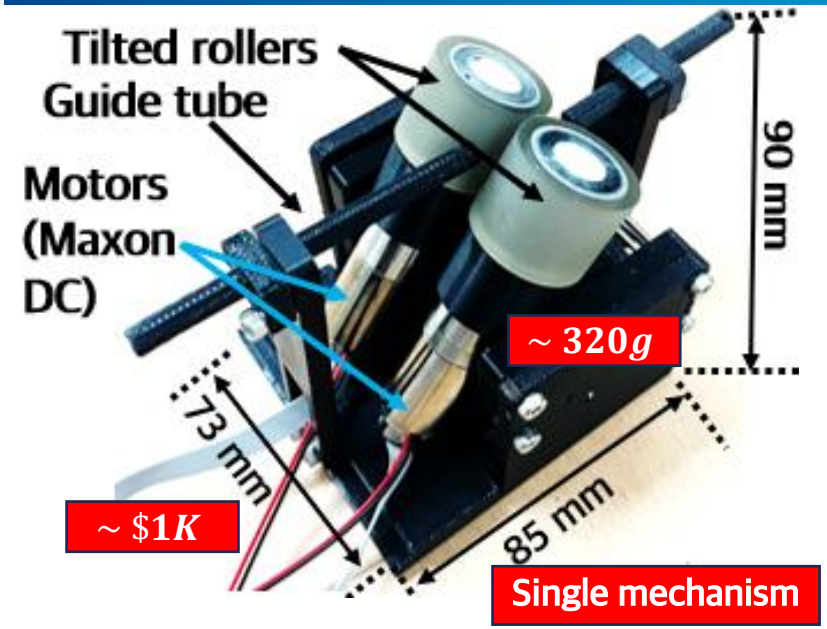
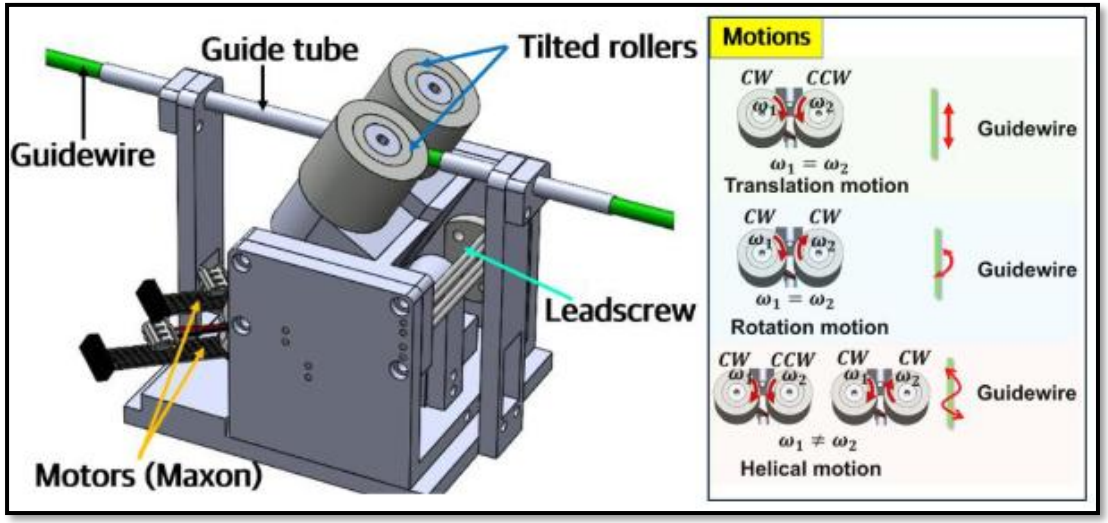
Multi-DOF Flexible End-Effector Actuator

A Dual Active Tilted Roller Actuator System (DATRAS)

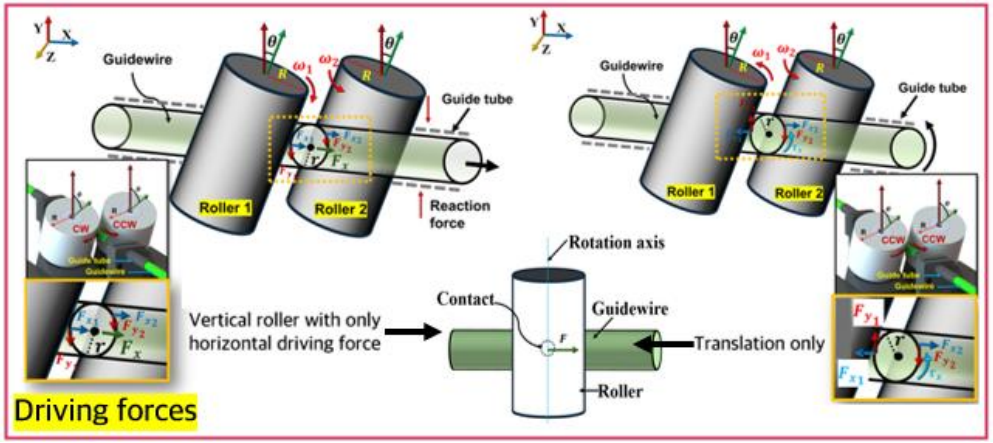


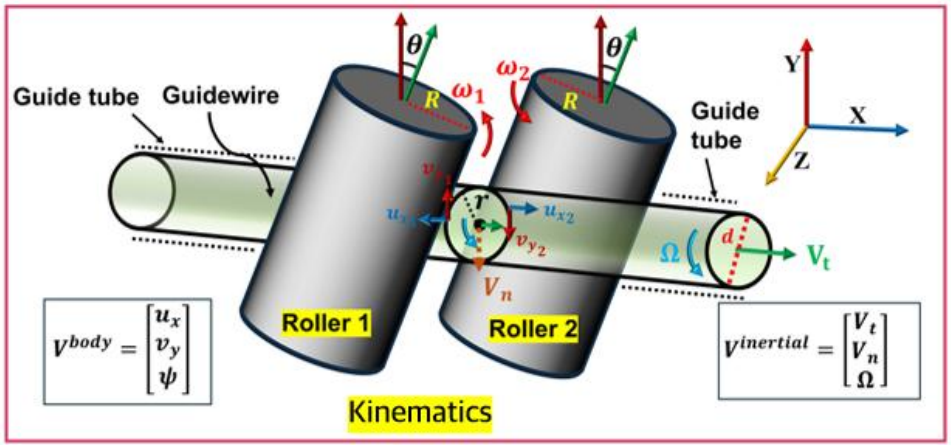


Motion: Translation, Rotation, Helical

$$Translation:\ F_x = F_{x_1}cos\theta + F_{x_2}cos\theta = \mu_s N_1 cos\theta + \mu_s N_2 cos\theta$$

$$Rotation:\ F_y = F_{y_1}sin\theta + F_{y_2}sin\theta = \mu_s N_1 sin\theta + \mu_s N_2 sin\theta$$



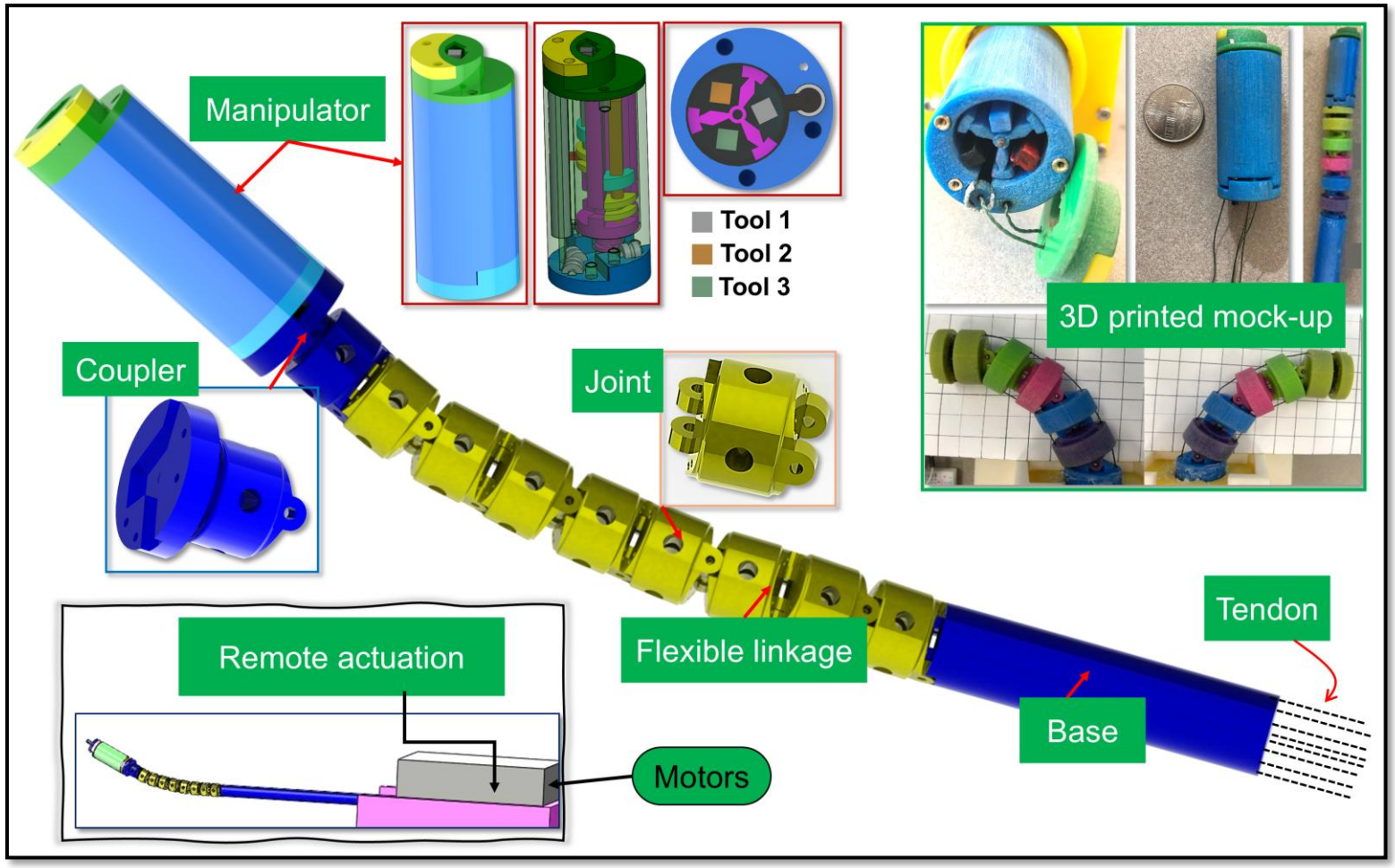


Tendon-driven Multi-functional Robotic Manipulator

II

Minimally Invasive Surgical Robot

Tendon-driven manipulator with flexible linkage mechanism:



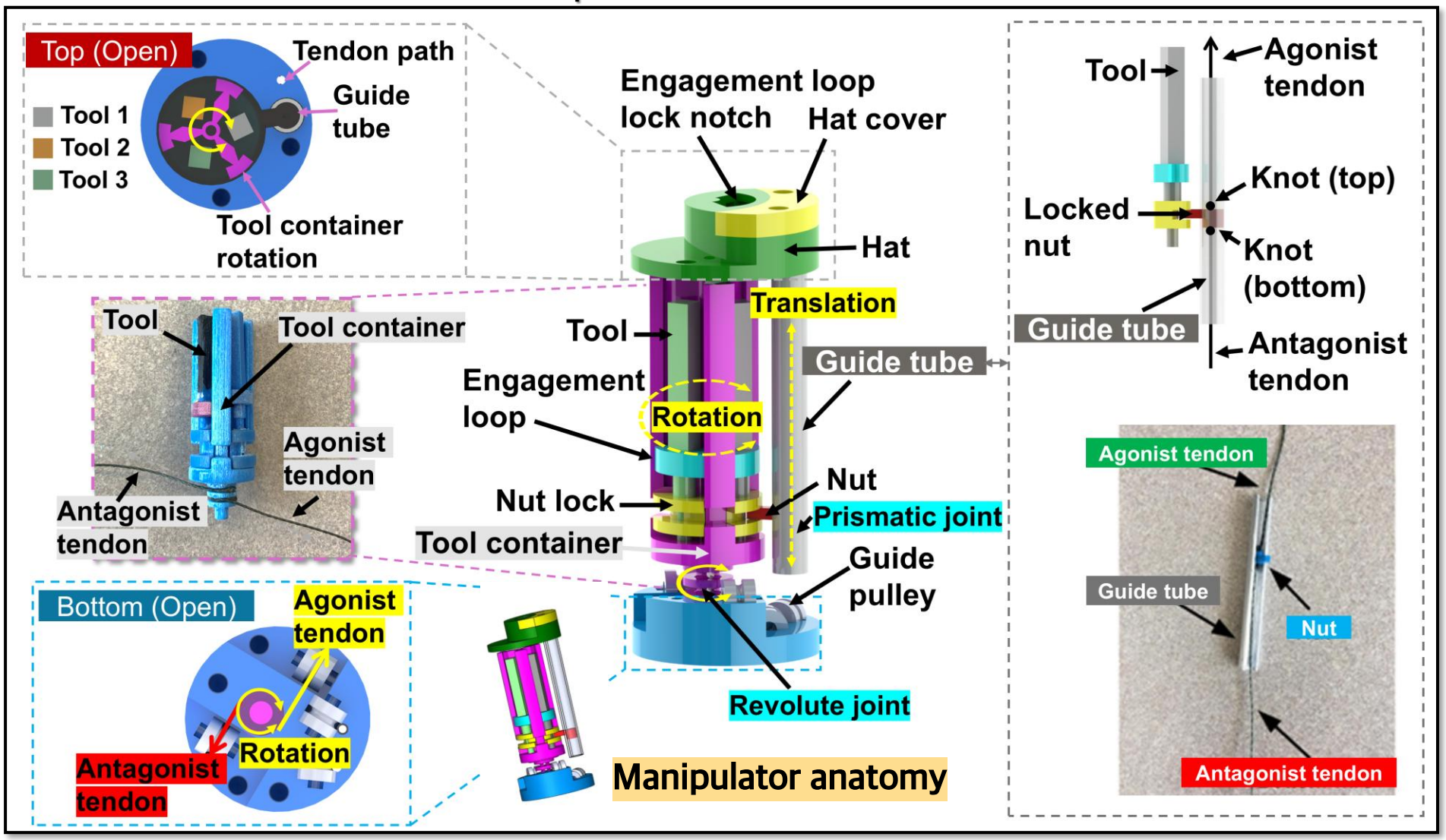
- Remote actuator placement
- Joint weight reduction
- Actuator protection
- Multiple tool lodging
- Faster tool changing
- Curvatures access

- Wire-actuated prismatic, rotational and universal joints mechanism

Tendon-driven Multi-functional Robotic Manipulator

Wire Actuated Manipulator Mechanism

Tendon-driven multi-functional manipulator mechanism:

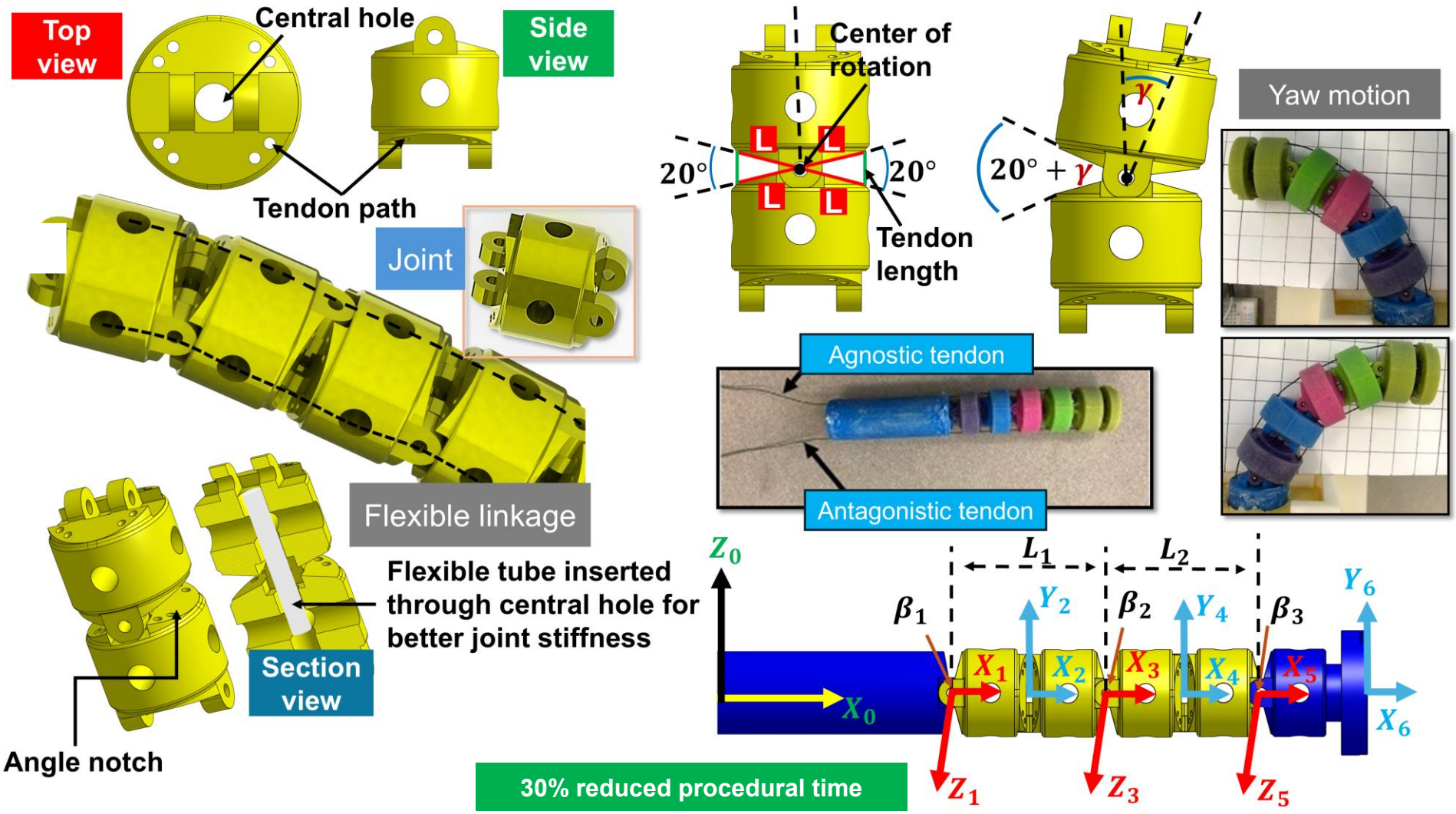


Tendon-driven Multi-functional Robotic Manipulator

II

Flexible Snake-like Robot Arm

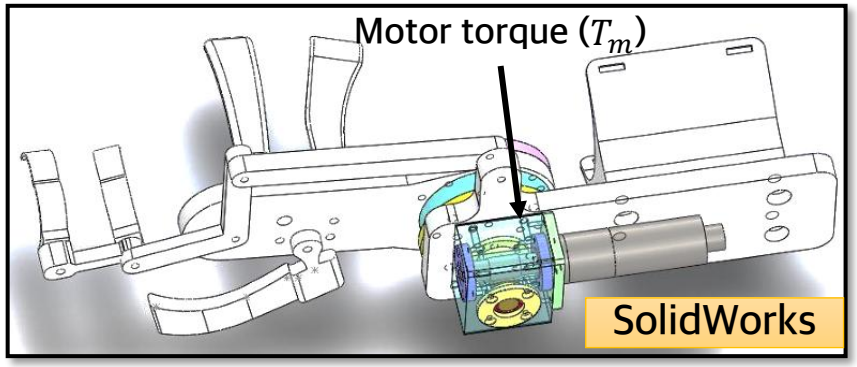
Tendon-driven flexible linkage mechanism:

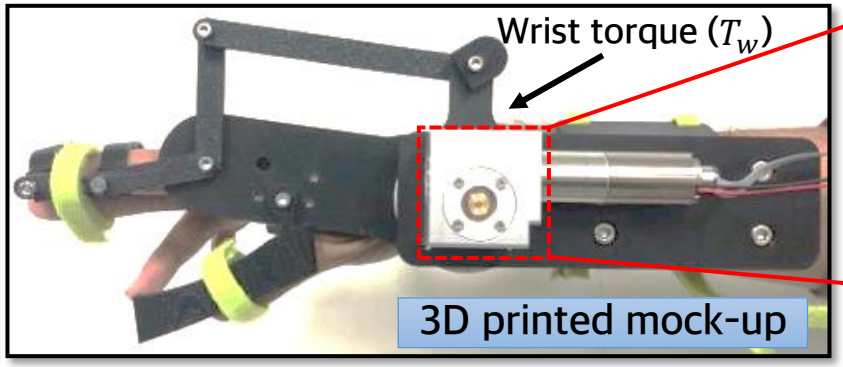


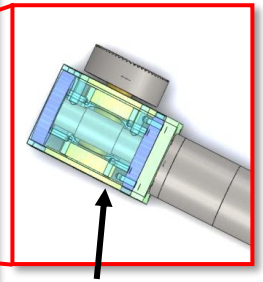
30% reduced procedural time

A 4-bar Mechanism Robotic Exoskeleton

Motor Augmented 4- Bar Linkage Mechanism

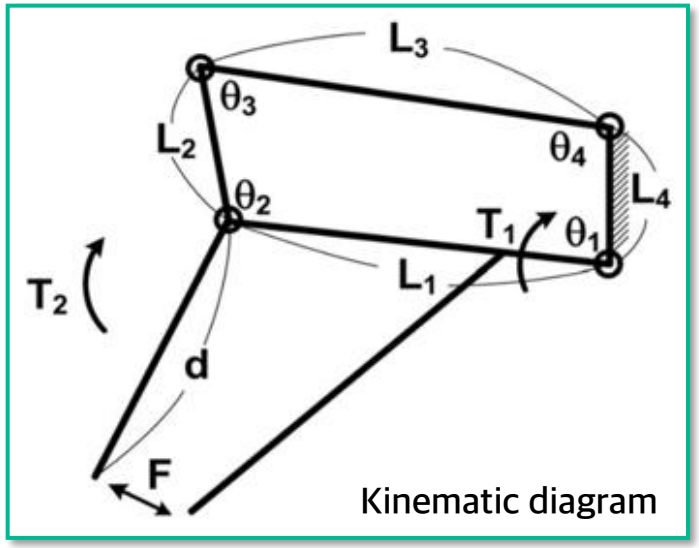


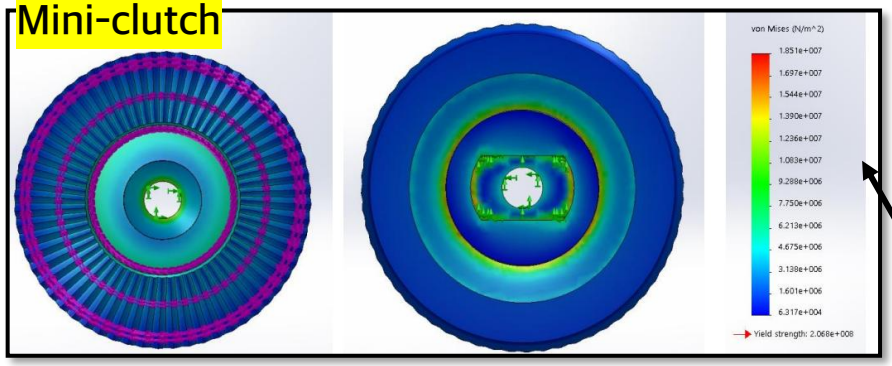




Worm gearbox with mini clutch

- On: Active
- Off: passive
- Includes safety mechanism





FEM structural analysis

- Structurally stable with maximum torque (1.38 Nm)
- Stainless steel: 24 mm diameter

Total grasp force:

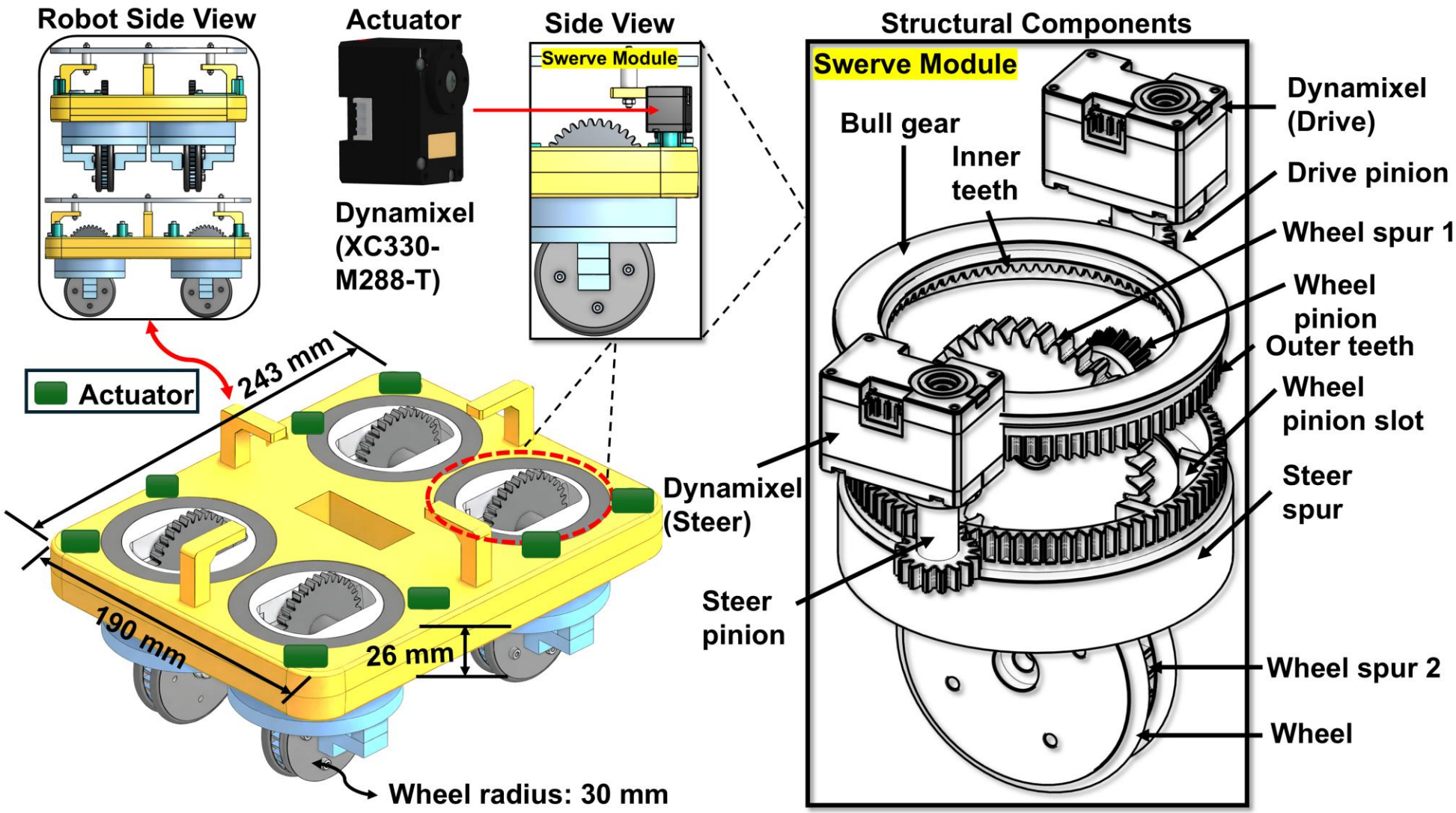
$$F = \frac{L_2 \cos(90 - \theta_3)}{L_2 \cos(90 - \theta_3) + L_1 \sin(\theta_1 + \theta_4)} \times \frac{1}{d}(T_m + T_w)$$

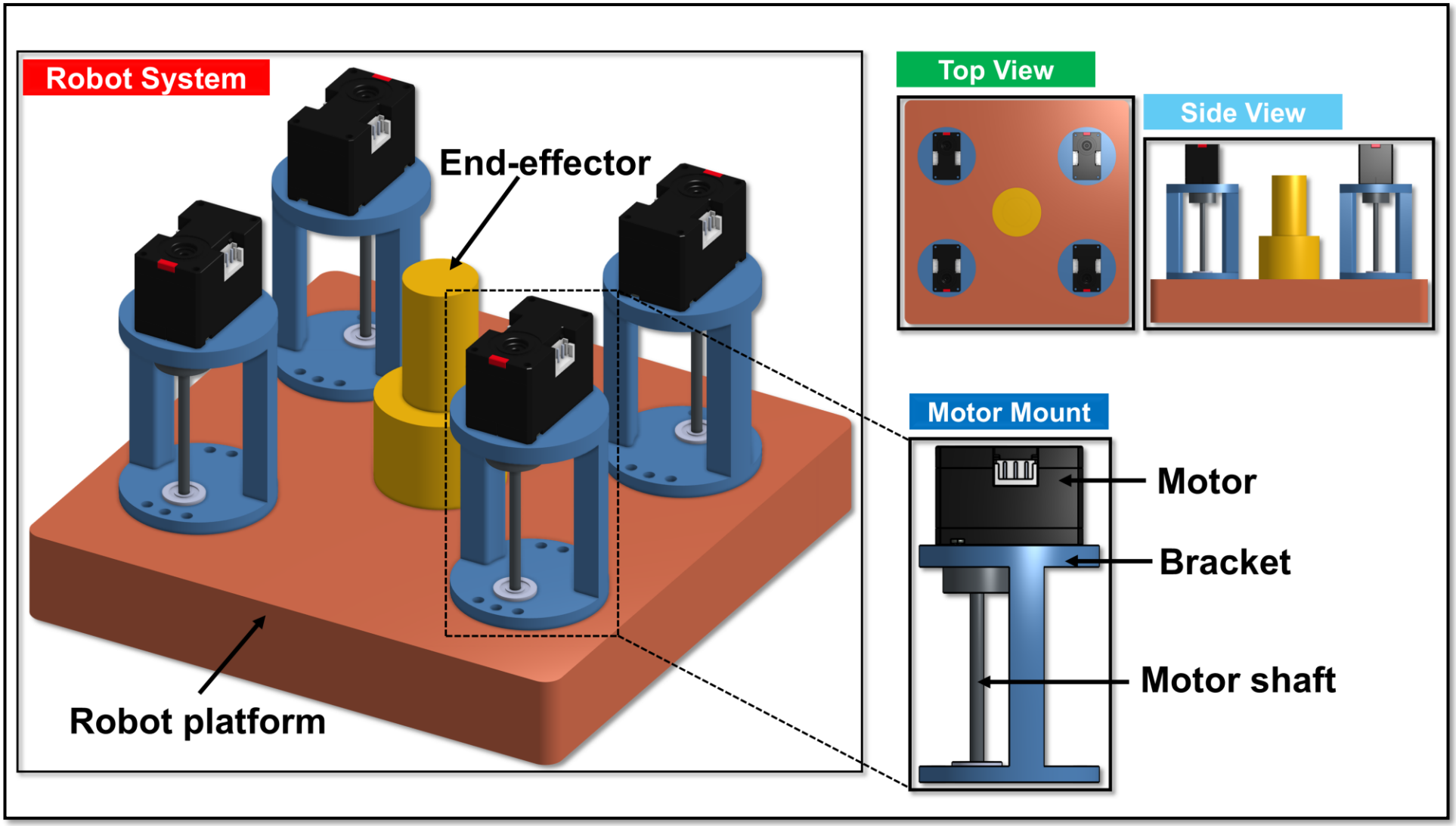
- Low-cost: single motor; no sensor
- Force tuning: active and passive mode
- Enables 75% daily tasks (15N force)
- Easy sanitization; modular design

Link lengths optimized through kinematics

Holonomic Mobile Robots

4 Wheel Swerve Drive Robot



Cable Driven Parallel Robot Mechanism

Cable Driven Gantry-type Cleaning Robot Mechanism